

# Hodge Theory on $T^4/\mathbb{Z}_3$ within FFGFT

Frobenius, Sector Pairing and Algebraic Cycles —  
a Known Special Case and Its Galois Structure

Johann Pascher  
ORCID 0009-0000-6518-4064

14 September 2026

## Contents

|                                                                        |   |
|------------------------------------------------------------------------|---|
| Hodge Theory on $T^4/\mathbb{Z}_3$ within FFGFT . . . . .              | 2 |
| 1 The Millennium Problem . . . . .                                     | 1 |
| 2 The Complex 2-Torus as an Abelian Surface . . . . .                  | 1 |
| 3 The Nine $\mathbb{Z}_3$ Fixed Points as Algebraic 0-Cycles . . . . . | 1 |
| 4 Frobenius Automorphism and Algebraic Frobenius . . . . .             | 2 |
| The Galois Frobenius on $\text{GF}(27)^*$ . . . . .                    | 2 |
| The Algebraic Frobenius on Étale Cohomology . . . . .                  | 2 |
| 5 Sector Pairing as Hodge Symmetry . . . . .                           | 2 |
| 6 The $\chi$ -Classes as Algebraic Basis . . . . .                     | 3 |
| 7 The Hodge Conjecture for $T^4/\mathbb{Z}_3$ . . . . .                | 3 |
| 8 What FFGFT Does Not Supply . . . . .                                 | 3 |
| 9 Summary . . . . .                                                    | 4 |

# **Hodge Theory on $T^4/Z_3$ within FFGFT**

### Abstract

The second Millennium Problem of mathematics — the Hodge Conjecture — asks whether every Hodge class of a smooth projective variety is a rational linear combination of algebraic cycle classes. For the base geometry of FFGFT, the orbifold  $T^4/\mathbb{Z}_3$ , this question admits a positive answer: the complex 2-torus  $T^4$  is polarisable in the form  $E_\omega \times E_\omega$  and hence an abelian surface for which the Hodge Conjecture is known to hold; the nine  $\mathbb{Z}_3$  fixed points furnish algebraic 0-cycles; and the sector pairing  $k \leftrightarrow -k$  (Doc. 343) is the Hodge symmetry  $H^{p,q} \leftrightarrow H^{q,p}$ . The Galois Frobenius  $x \mapsto x^3$  on  $GF(27)^*$ , by contrast, is not identical to the algebraic Frobenius on étale cohomology: the latter's eigenvalues on  $H^1$  have absolute value  $\sqrt{3}$  and are not roots of unity. The Millennium Problem in its general form — for arbitrary smooth projective varieties — remains open. What FFGFT supplies is structural completeness for its own geometry.

## Status Markers

|                  |                                                       |
|------------------|-------------------------------------------------------|
| <b>[B]</b>       | Algebraically proved.                                 |
| <b>[K]</b>       | Numerically confirmed.                                |
| <b>[S]</b>       | Structurally plausible; physical identification open. |
| <b>[E]</b>       | Established external mathematics.                     |
| <b>[Q]</b>       | Taken from the literature, correctly cited.           |
| <b>[SETZUNG]</b> | Axiomatic stipulation.                                |

## 1 The Millennium Problem

The Clay Mathematics Institute states the Hodge Conjecture as follows **[Q]**: On a non-singular projective algebraic variety  $X$  over  $\mathbb{C}$ , any Hodge class — that is, any cohomology class of type  $(k, k)$  — is a rational linear combination of classes of algebraic cycles of codimension  $k$ .

More precisely: the Hodge decomposition

$$H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X)$$

splits cohomology by type; Hodge classes are the elements of  $H^{k,k}(X) \cap H^{2k}(X, \mathbb{Q})$ . The conjecture states that these classes are generated by genuine geometric objects — subvarieties whose cohomology classes are rationally combined. For smooth projective curves and surfaces in bidegree  $(1, 1)$  this follows from the Lefschetz theorem on  $(1, 1)$ -classes **[E]**. In general dimension the conjecture is open.

## 2 The Complex 2-Torus as an Abelian Surface

The real geometry  $T^4 = S^1 \times S^1 \times S^1 \times S^1$  carries in FFGFT the structure of a *complex 2-torus*: two complex coordinate pairs  $(z_1, z_2) \in \mathbb{C}^2/\Lambda$ , where  $\Lambda$  is the mode lattice **[SETZUNG]** (Doc. 001).

A polarisable complex 2-torus — here  $E_\omega \times E_\omega$  — is an abelian surface, an abelian variety of complex dimension 2. By the Riemann conditions  $\mathbb{C}^2/\Lambda$  is projective only with a polarisation; the generic 2-torus is not algebraic, whereas  $E_\omega \times E_\omega$  as a product of elliptic curves is. For abelian varieties the Hodge Conjecture is known in the cases relevant here **[E]** **[Q]**:

- For abelian surfaces (complex dimension 2) the conjecture holds completely for all Hodge classes:  $H^{1,1}$  is covered by the Lefschetz theorem,  $H^{2,2}$  is generated by the point class.
- Products of elliptic curves and tori over  $\mathbb{Q}$  are fully spanned by algebraic cycles.

**[B]** *Theorem A*:  $T^4$  as the abelian surface  $E_\omega \times E_\omega$  satisfies the Hodge Conjecture: all  $(k, k)$ -classes are rational linear combinations of algebraic cycles.

Proof: In complex dimension 2,  $(k, k)$ -classes exist only for  $k = 0, 1, 2$  (Hodge diamond  $h^{p,q} = \binom{2}{p}\binom{2}{q}$ ,  $h^{1,1} = 4$ ). For  $k = 0$  and  $k = 2$ : trivial via the fundamental class ( $H^{0,0}$ ) and the point class ( $H^{2,2}$ ), respectively. For  $k = 1$ : Lefschetz theorem on  $(1, 1)$ -classes (known **[E]**).  $H^{2,2}(T^4)$  is moreover generated by products of  $(1, 1)$ -classes, hence algebraic.  $\square$

## 3 The Nine $\mathbb{Z}_3$ Fixed Points as Algebraic 0-Cycles

Under the  $\mathbb{Z}_3$  action on  $T^4$  there are exactly nine fixed points (Doc. 001/330) **[B]**. These fixed points are algebraic 0-cycles — points on the variety  $T^4/\mathbb{Z}_3$  with well-defined cohomology classes.

**[B] Theorem B:** The nine  $\mathbb{Z}_3$  fixed points of  $T^4/\mathbb{Z}_3$  are algebraic 0-cycles of codimension 4. Their cohomology classes generate the twisted sector of the orbifold cohomology. In the orbifold cohomology (Chen–Ruan) **[Q]**:

$$H_{\text{CR}}^*(T^4/\mathbb{Z}_3) = H^*(T^4)^{\mathbb{Z}_3} \oplus \bigoplus_{9 \text{ fixed points}} H^*(\{p\})$$

The second summand is algebraic: every point is an algebraic 0-cycle; its cohomology class is trivially algebraic. **[E]**

Under the Witt-pair structure of Doc. 341:

- **Even indices 0,2,4,6** =  $\mathbb{Z}_3$  fixed points = algebraic 0-cycles **[B]**
- **Odd indices 1,3,5,7,8** = three-cycles = higher-dimensional algebraic cycles **[B]**

This is literally the Hodge statement: topological class = algebraic cycle, for all classes in the twisted sector.

## 4 Frobenius Automorphism and Algebraic Frobenius

This section separates the Galois Frobenius on  $GF(27)^*$  from the algebraic Frobenius of étale cohomology.

### The Galois Frobenius on $GF(27)^*$

Doc. 339 proves **[B]**: the Frobenius automorphism on  $GF(27) = GF(3^3)$  is  $x \mapsto x^3$ . It acts on  $GF(27)^* \cong \mathbb{Z}_{26}$  and generates the sector structure: fixed points  $\{+1, -1\}$  (massive sector), eight three-cycles (gluons/massless sector).

### The Algebraic Frobenius on Étale Cohomology

In the arithmetic geometry of algebraic geometry **[Q]**, for a variety  $X$  over  $GF(q)$  the geometric Frobenius is  $F_q : x \mapsto x^q$  on points. It induces an endomorphism  $F_q^*$  on the  $\ell$ -adic cohomology  $H_{\text{ét}}^i(X, \mathbb{Q}_\ell)$ . The eigenvalues of  $F_q^*$  are algebraic numbers that, by the Weil–Deligne theorem **[E]**, determine the zeta function and the Betti numbers of  $X$  (weights:  $|\alpha| = q^{i/2}$  on  $H^i$ ). The Hodge numbers  $h^{p,q}$  are not determined by them.

*Theorem C (no identification):* The Galois Frobenius  $x \mapsto x^3$  on  $GF(27)^*$  is not identical to the algebraic Frobenius on  $H_{\text{ét}}^1$  of  $T^4/\mathbb{Z}_3$  over  $GF(3)$ . The obvious identification fails because, by Weil, the Frobenius eigenvalues on  $H^1$  of an abelian surface over  $\mathbb{F}_q$  have absolute value  $\sqrt{q}$ , here  $\sqrt{3} \approx 1.732$ : they are four Weil numbers, not roots of unity and not 26 of them. Example:  $y^2 = x^3 - x$  over  $GF(3)$  has  $\#E = 4$  and eigenvalues  $\pm i\sqrt{3}$  (cross-check  $\#E(GF(27)) = 28$ ); for  $E \times E$  there are four eigenvalues with  $\alpha^{26} = -3^{13}$ . An identification with the elements of  $GF(27)^*$  does not follow. The two share only the map  $x \mapsto x^3$  as generator of  $\text{Gal}(GF(27)/GF(3))$ .

**[S] Remark:** The physical interpretation — Frobenius orbits as particle sectors — is an FFGFT classification and not part of the mathematical proof.

## 5 Sector Pairing as Hodge Symmetry

Doc. 343, Theorem D **[B]** proves: the sector pairing  $k \leftrightarrow -k \pmod{13}$  on the primitive  $GF(27)^*$  classes pairs  $f_1 \leftrightarrow f_2$  and  $f_3 \leftrightarrow f_4$ .

**[B] Theorem D:** This sector pairing is the algebraic realisation of the Hodge symmetry  $H^{p,q} \leftrightarrow H^{q,p}$  under complex conjugation.

Proof: The Hodge decomposition on  $T^4$  has  $H^{1,0}$  and  $H^{0,1}$  as a conjugate pair, and  $H^{2,1}$  and  $H^{1,2}$  likewise. Complex conjugation  $\bar{\cdot} : H^{p,q} \rightarrow H^{q,p}$  is the algebraic datum selecting real cohomology classes. On  $GF(27)^*$  the corresponding map is inversion  $g^k \mapsto g^{-k}$ , i.e. negation of the exponent in  $\mathbb{Z}_{26}$ . It must be distinguished from field negation  $e \mapsto -e = e \cdot g^{13}$ : that map sends every order-26 orbit to an order-13 orbit and is therefore not a pairing within the primitive classes (check script pruef\_363b, part L3). Inversion, by contrast, is an involution on the four primitive classes with no fixed orbit; it reduces  $\{f_1, f_2, f_3, f_4\}$  to two real classes — exactly the Hodge symmetry.  $\square$

From Doc. 343 Theorem D''' **[B]**:  $\{f_3, f_4\}$  is the Dirac layer (charged leptons),  $\{f_1, f_2\}$  the Majorana layer (neutrinos). The Hodge pairing separates exactly the physical sectors.

## 6 The $\chi$ -Classes as Algebraic Basis

Doc. 342 **[B]** classifies the eight irreducible cubic polynomials over  $GF(3)$ :

- Four primitive polynomials of order 26:  $f_1, f_2, f_3, f_4$
- Four polynomials of order 13:  $g_1, g_2, g_3, g_4$

**[B] Theorem E:** The eight polynomials generate the factorisation classes of  $GF(3)[x]/(x^{26}-1)$  completely (Doc. 342).

A reading of these classes as a  $\mathbb{Q}$ -basis of algebraic cycles on  $T^4/\mathbb{Z}_3$  is not established thereby: it would require the irreducible factors of the characteristic polynomial of  $F_3^*$  on  $H^i$  to be the eight polynomials of Doc. 342, i.e. the identification of Theorem C, which does not hold.

## 7 The Hodge Conjecture for $T^4/\mathbb{Z}_3$

**[B] Main Theorem:** All Hodge classes on  $T^4/\mathbb{Z}_3$  are rational linear combinations of algebraic cycles.

Proof from Theorems A and B:

1.  $H^*(T^4)^{\mathbb{Z}_3}$  (untwisted sector): All Hodge classes come from the  $\mathbb{Z}_3$ -invariant part of  $H^*(T^4)$ . Since  $T^4$  as an abelian surface satisfies the Hodge Conjecture (Theorem A) and the  $\mathbb{Z}_3$  projection maps algebraic cycles to algebraic cycles, all invariant Hodge classes are algebraic.
2. Twisted sector: The nine fixed points are algebraic 0-cycles (Theorem B); their cohomology classes are trivially algebraic.
3. By the Chen–Ruan decomposition (Section 3), orbifold cohomology consists of exactly these two summands; no Hodge class lies outside them. The Frobenius structure is not needed for this.  $\square$

*In plain language:* For the specific geometry of FFGFT the Hodge Conjecture is not an open problem. It follows from combining the abelian-surface theorem and orbifold cohomology. The Millennium Problem in its general form is not solved by this.

## 8 What FFGFT Does Not Supply

**General varieties.** The Hodge Conjecture is open for smooth projective varieties of arbitrary dimension.  $T^4/\mathbb{Z}_3$  is a very special class — a torus orbifold with rich algebraic structure. For a general variety of Hodge type (2, 2) in dimension 4 none of the above arguments apply.

**No singularities in the FFGFT reading.** FFGFT does not work on the quotient space  $T^4/\mathbb{Z}_3$  as a point set, but on the smooth  $T^4$  with  $\mathbb{Z}_3$  as a lattice symmetry of the modes (Doc. 314: geometry  $\rightarrow$  spectrum + fibre; the orbifold is a lattice symmetry, not an additional assumption). The cohomology is the  $\mathbb{Z}_3$ -equivariant cohomology of a smooth abelian surface  $E_\omega \times E_\omega$ ; Theorem A applies directly, with no detour through singularities. **[B]**

If one nevertheless forms the quotient space, the  $\mathbb{Z}_3$  action has, by Doc. 330, eigenvalues  $\{\omega, \omega^2\}$  on  $\mathbb{C}^2$ , hence  $\det = 1$ : it lies in  $SU(2)$ . The nine fixed points are then  $A_2$  points and admit a crepant resolution; the result is a K3 surface — smooth, projective, and for K3 surfaces the Hodge Conjecture is known (Lefschetz on  $H^{1,1}$  suffices in dimension 2) **[E]**. In neither reading is there an obstruction.

**Premise and level of representation.** Higher-dimensional spaces and objects may be constructed freely as long as they are translations of the  $T^4$  content (Type III, Doc. 270/330 **[K]**): they contain no more than  $T^4$ , and no projection from them can yield more than the mode content of  $T^4$  (Doc. 314: lattice = index set of Fourier modes; Doc. 343: the resolution limit lies in the object itself). All theorems of this document are such translations — the same compact geometry in the Galois language of  $GF(27)^*$  and in the language of the Hodge decomposition, bijective and lossless. The document contains no Type I or Type II operation (duality decompactification, geometric projection) and therefore makes no claim about the observability of individual cycles; that is governed by the projection theory (Doc. 285, 330, 333).

**Non-commutative extension.** Orbifold cohomology (Chen–Ruan) agrees with the classical Hodge decomposition in the untwisted sector; the twisted sector has a grading shift modifying the Hodge symmetry. The statement here holds for the classical picture; the Chen–Ruan extension is to be rated **[S]**.

## 9 Summary

| FFGFT structure                         | Hodge<br>dence                                                                                        | correspon- | Doc.    | Status     |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|------------|---------|------------|
| $T^4$ as abelian 2-torus                | Hodge<br>known                                                                                        | conjecture | 001     | <b>[E]</b> |
| 9 $\mathbb{Z}_3$ fixed points           | Algebraic 0-cycles                                                                                    |            | 001/330 | <b>[B]</b> |
| Frobenius $x \mapsto x^3$ on $GF(27)^*$ | No identification with the Frobenius on $H_{\text{et}}^1$ (eigenvalues of absolute value $\sqrt{3}$ ) |            | 339/341 | —          |
| Sector pairing $k \leftrightarrow -k$   | Hodge<br>$H^{p,q} \leftrightarrow H^{q,p}$                                                            | symmetry   | 343     | <b>[B]</b> |
| $\chi$ -classes (8 polynomials)         | Factorisation classes of $GF(3)[x]/(x^{26}-1)$ ; no cycle basis                                       |            | 342     | <b>[B]</b> |
| Hodge conjecture for $T^4/\mathbb{Z}_3$ | All Hodge classes algebraic                                                                           |            | —       | <b>[B]</b> |
| General Hodge conjecture                | Millennium Problem open                                                                               | Problem    | —       | —          |

The Hodge Conjecture is not an open problem for  $T^4/\mathbb{Z}_3$ . It follows from Theorem A and orbifold cohomology. Of the Galois structure of  $GF(27)^*$ , the sector pairing realises the Hodge symmetry (Theorem D); the Galois Frobenius and the  $\chi$ -classes, by contrast, cannot be identified with the Frobenius or the cycles of étale cohomology (Theorem C). The general Millennium Problem is unaffected.